(iii) The maximum grade the truck can negotiate at the above engine speed in second gear. 5

(iv) The maximum drawbar pull available on level at the above engine speed in second gear. 3

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4. (a) What is the function of a brake in an automobile? 2

(b) What is the necessity of universal joint, slip (sliding) joint and a differential in a motor vehicle? 7

(c) Describe with the help of neat sketches the working of the hotchkiss drive and the hydraulic brake. 7

(d) Draw the cross-section of a truck tyre, lable the nomenclature of various parts. What is the reason for excessive type wear? 7

5. (a) Define camber angle. 2

(b) Write notes on any three of the following : 14

(i) Ackerman steering and condition for true rolling on curves (Mathematical derivation)

(ii) Slip angle, toe-in and castor angle

(iii) Understeer and oversteer

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(iv) Construction and working of 12 volt battery

(v) Lighting circuit of a car

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B. E. (Seventh Semester) Examination, April-May 2016

(New Scheme)

(Mech. Branch)

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AUTOMOBILE ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

***Note** : Attempt all five questions. Part (a) of each question is compulsory. Attempt for sixteen marks only from each question. Due credit will be given to neat sketches.*

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1. (a) What is the function of damper? 2

(b) List the various components of a chasis. Draw the layout showing various compoenent of chasis. 7

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(c) What is the purpose of independent suspension? Draw the neat sketch and explain the common type of independent suspension system used in cars at the front end. 7

(d) Draw a neat sketch of a hydraulic shock absorber and explain its construction and working. 7

2. (a) What is the function of a clutch? 2

(b) Determine the size of the clutch plate suitable for a car employing a single plate type of friction clutch and developing 37·5 kW at 4200 r.p.m. The inside diameter of the clutch plate is 0·6 times its outside diameter and it is to be ensured that even after a loss of 30% of the engine torque due to wear of the clutch facing, the clutch does not slip. The intensity of pressure on the facing is not to exceed 70 kPa. Assume $\mu = 0{\cdot}3$. 7

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(c) What are the differnece between : 7

(i) Dry clutch and wet clutch

(ii) Clutch plate and pressure plate

(iii) Clutch and fluid flywheel

(d) Explain by means of neat sketches the construction and working of a fluid flywheel and describe clearly the principle of torque transmission in the same. 7

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3. (a) What is the purpose of transmission in an automobile? 2

(b) (i) With the help of neat sketch, explain the construction and working of a constant mesh gear box. What as its merits? 7

(ii) With the help of neat sketches explain the construction and working of torque convertor. 7

Or

The coefficient of rolling resistance for a truck weighing 6500 kg is 0·017 and the co-efficient of air resistance is 0·003 in the formula $R = KrW + KaAV^2$ kg, where A is m^2 of frontal area and V the speed in Km/hr. The transmission efficiency in top gear of 6 : 1 is 90% and that in the second gear of 14 : 1 is 80% the frontal area is 5·4 m^2. If the truck can have a maximum speed of 80 km/hr, in top gear. Calculate :

(i) The engine b.h.p. required 3

(ii) The engine speed, its the driving wheel have an effective diameter of 82 cm. 3

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